Journal of The
Electrochemical Society

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To cite this article: Feiyun Cui *et al* 2020 *J. Electrochem. Soc.* **167** 037525

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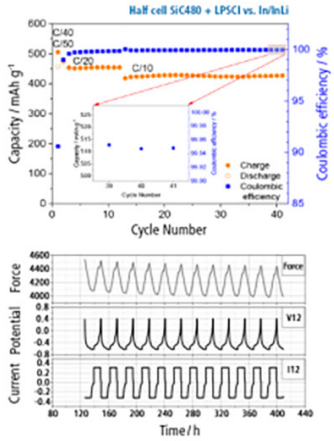

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Journal of The Electrochemical Society, 2020 **167** 037525

Review—Measurement and Analysis of Cancer Biomarkers Based on Electrochemical Biosensors

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Cancer is a dreadful disease with a high mortality rate, and it has become more and more prevalent worldwide. Early diagnosis, prognosis and treatment monitoring with robust and non-invasive tools will potentially be the future focus. Electrochemical biosensor can be a strong candidate for cancer theranostics owing to their advantage of ultra-sensitivity, high selectivity, low cost, quick readout, and simplicity. Furthermore, electrochemical biosensors are easier to be miniaturized and mass fabricated, which grant them a better fit for point-of-care applications. In this review, various electrochemical measurement methods, bioreceptor surface, signal generation and amplification, integration of electrochemical sensors in microfluidic chips were summarized. Especially, multiplexed and ratiometric electrochemical biosensor were emphasized in cancer biomarkers detection. Then, measurement and analysis of cancers based on electrochemical biosensors in molecular level (DNA, RNA, and protein), organelle level (exosomes), cell level (cell counting, phenotypic and metabolism analysis, drug sensitivity monitoring) were comprehensively discussed. As a new research trend, the integration of electrochemical biosensors in cancer-on-a-chip has been highlighted. In brief, we present an overall review of current advances in cancers measurement and analysis using electrochemical biosensors. Finally, the current challenges and future directions were discussed.

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Manuscript submitted September 24, 2019; revised manuscript received November 15, 2019. Published December 19, 2019. *This paper is part of the JES Focus Issue on Sensor Reviews.*

Cancer is a great threat to our life. In 2019, as American Cancer Society estimated, 1,762,450 new cancer cases and 606,880 cancer deaths are projected to occur in the United States.[1] In other words, about 4,829 people are diagnosed with cancer and about 1,663 people died of cancer every day. Effective early detection and treatment strategies can be greatly helpful to reduce cancer incidence rate and mortality. Current tumor diagnosis relies on a variety of complex clinical settings, which include X-ray, computerized tomography (CT), magnetic resonance imaging (MRI), positron emission tomography, endoscopy, sonography, thermography, cytology and biopsy. Besides that, molecular tools based on both genomic and proteomic are increasingly used, such as polymerase chain reaction (PCR), enzyme linked immunosorbent assay (ELISA), radioimmunoassay (RIA), immunohistochemistry (IHC), and flow cytometry.[2–4] The existing technologies and methods are efficient, however, most of them are invasive, expensive, time-consuming, and limited to laboratory centers in large hospitals. Besides high morbidity and mortality, another serious problem of cancer is the widening of the socioeconomic inequalities, with the most notable gaps for the most preventable cancers. For example, compared with the most affluent counties, mortality rates in the poorest counties were 2-fold higher for cervical cancer and 40% higher for male lung and liver cancers during 2012–2016.[1] Therefore, development of low cost, non-invasive tools for cancer theranostics are urgently needed.

Biosensors have the potential to be alternative tools since they can provide fast, accurate, sensitive early detection; monitor cancer metastasis and angiogenesis; measure and analyze effectiveness of anticancer chemotherapy drugs in a non-invasive style.[5] Among various biosensors,[6,7] electrochemical biosensors possess the advantages of low cost, simplicity, easier to be miniaturized and mass fabricated. They also can be used as point-of-care (POC) devices at home or doctor's office. Therefore they have received significant attention and extensive efforts have been devoted to developing ultrasensitive electrochemical biosensors for detection of cancer markers with high selectivity.[8,9] These efforts include discovery of the bioreceptors with high specificity and affinity, design of novel redox tags to conduct multiplex bioassay and ratiometric electrochemical assay, development of signal amplifiers based on nanomaterials, and integration of electrochemical biosensors with microfluidic chips. The advantages, component parts, and various electrochemical measurement methods are presented in Fig. 1A. Cancers are normally induced by uncontrollable cell proliferation which is accompanied by a series of cellular events, such as specified genomic alterations in the normal cell (initiation), following transcriptome and protein alteration, tumor formation, cancer cell migration, and tumor metastasis. The size, phenotype, and metabolism of cancerous cells usually change compared to normal cells. Meanwhile, the tumor related microenvironment also changes.[8,10] Immune cells secrete some special cytokines in response to cancer cells.[11–14] The extracellular matrix (ECM) also undergoes significant changes under the action of matrix metalloproteinases (MMPs).[15] Throughout the process of tumor formation, development and metastasis, many different cancer biomarkers are produced, such as altered genomic, circulating tumor DNA (ctDNA), microRNA, cancer antigens, cytokines, MMPs, exosomes, Circulating Tumor Cells (CTCs) etc.[8,16,17] The biomarkers originating from tumorigenesis, migration and colonization to a second site are systematically presented in Fig. 1B.

This review provides an overall view of recent progresses (mainly focusing on reference reported in years of 2014∼2019) achieved in electrochemical biosensors for cancer biomarkers detection and analysis in different stages. The emphasis is placed on detection strategy and performance of these biomarkers. Discussions are focused mainly on usage of various electrochemical detection methods, redox species and signal amplification strategies. As a new research trend, the integration of electrochemical biosensors in cancer-on-a-chip has been highlighted. In the last part, challenges and possible solutions of applying electrochemical biosensors to clinical application are proposed.

Electrochemical Biosensor

***Various electrochemical measurement methods.*—Amperometry** is the most often employed technique which continuously measures current resulting from the oxidation or reduction of an electroactive species in a biochemical reaction motivated at a constant applied potential. It is widely used for cancer biomarker and cancer cell detection. For instance, Annexin II and MUC5AC, the profound diagnostic biomarkers in lung cancer have been detected by Kim and coworkers using amperometric immunosensing methods with the Limit Of Detection (LOD) of 280 ± 8.0 pg mL^{-1}.[18] In another example, a selective and sensitive biosensing prototype based on graphene nanocomposite with functionalized AuNPs has been reported for the detection of human cervical cancer biomarker miR-21.[19] The CTCs are also targeted using amperometric techniques. For example, Moscovici et al. have reported rapid CTC detection,[20] where

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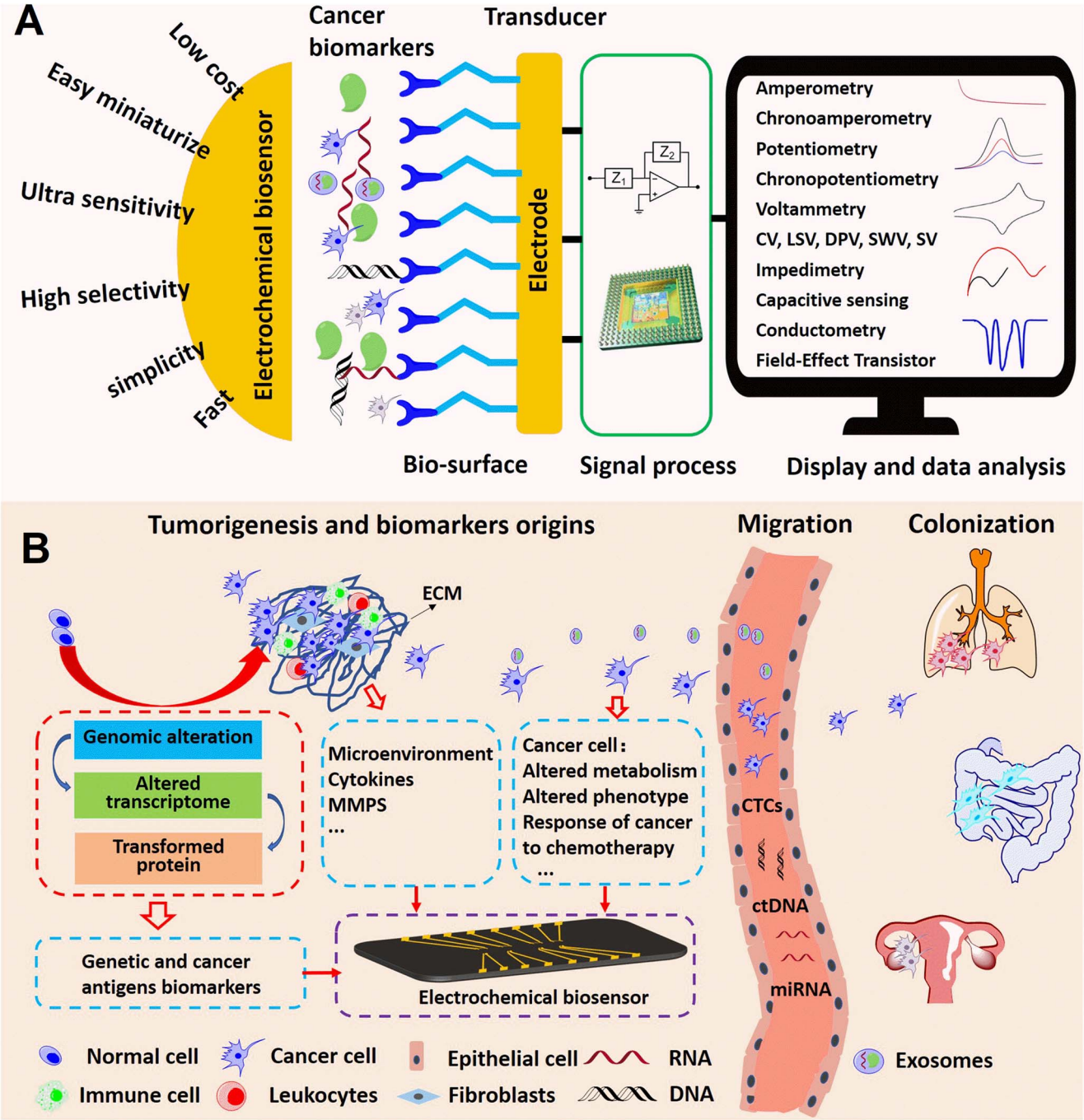

Figure 1. Schematic diagram of advantages, component parts, and various measurement methods of electrochemical biosensor (A) and the cancer biomarker originating from tumorigenesis, cancer cell migration and colonization to a second site (B).

circulating prostate cancer cells were detected using gold array electrodes. In another work, Chandra et al.[21] have developed an amperometric biosensor for the detection of cancer cells using antipermeability glycoprotein functionalized nanoprobe, which has shown Linear Range (LR) and LOD of 50 to 10^5 cells mL^{-1} and 23 ± 2 cells mL^{-1}, respectively.

Another amperometric technique is known as **chronoamperometry**, where a square-wave potential is applied to the working electrode and a steady state current is measured as a function of time. Barhoumi et al. applied chronoamperometric immunosensor to detection tumor necrosis factor-α (TNF-α), which showed good performance within the clinically relevant concentration range with a precision of 8% and a LOD of 0.3 pg mL^{-1}.[22] Tsai et al. demonstrated that chronoamperometric method can be used as an alternative method for rapidly assessing the viability of MDA-MB-231breast cancer cells.[23]

Potentiometry devices allow the quantification of an electrical potential difference between two electrodes when the cell current is zero. Potentiometric approaches were also used for cancer cell and biomarker monitoring. For potentiometric measurements, the relationship between the concentration and the potential is governed by the Nernst equation. It offers low LOD (between 10^{-8} and 10^{-11} M) which is important for cancer detection as the concentration of biomarkers are very low in the early stages.[24] Jia and coworkers have reported a Light-Addressable Potentiometric Sensor (LAPS) for the detection of human phosphatase of regenerating liver-3 (hPRL-3), a prognostic biomarker of liver cancer. In this work, hPRL-3 and mammary adenocarcinoma

cells (MDA-MB-231) have been detected with LR of 0.04−400 nM and 0–10^5 cells mL^{-1}, respectively.[25] In another work, carcinoembryonic antigen (CEA) was detected from CEA-producing LoVo human colon cancer cells using surface molecular imprinted self-assembled monolayers (SAM) of hydroxyl-terminated alkanethiol and template biomolecules on gold-coated silicon chip as bio-recognition element for a potentiometric biosensor with reported LR of 2.5–250 ng mL^{-1}.[26] In malignant pleural mesotheliomas (MPM), the overexpressed proteinaceous biomarker hyaluronan-linked protein 1 (HAPLN1) has also been targeted for using label-free potentiometric detection and achieved the LOD in pM range with a quick response time of 2–5 min in real sample.[27] In a study carried out by Goda and coworkers, a hybridization-based potentiometric microarray was developed to detect the exosomal miRNA.[28] Furthermore, cancerous cells have also been targeted using potentiometric techniques to study the electrochemistry of proximal micro-environments, since the cancerous cells usually release lactate which introduces the fluctuation of pH of media. Based on this concept, Shaibani and coworkers targeted cancer cells (MDA-MB-231) with the LOD of 10^3 cells mL^{-1}. Also, this work confirmed changes in pH flux surrounding the neoplasm consisting cancer cells which infers the correlation to the metabolism of altered cells.[29] Similarly, anti-EpCAM functionalized graphene oxide potentiometric biosensor was developed for the selective detection of CTCs of prostate cancer using the LAPS technique.[30]

In **chronopotentiometry** the potential is measured as a function of time in response to a constant or square-wave current. Chronopotentiometric stripping (CPS) was used by Belicky et al. to investigate the prostate specific antigen (PSA) and its interactions with lectins capable to recognize PSA glycans occurring in healthy people or in patients with prostate cancer.[31]

Voltammetry belongs to a category of electro-analytical methods, through which information about an analyte is obtained by varying a potential and then measuring the resulting current. Since there are many ways to vary a potential, there are also many forms of voltammetry methods i.e. **cyclic voltammetry (CV), linear sweep voltammetry (LSV), differential pulse voltammetry (DPV), square wave voltammetry (SWV), stripping voltammetry (SV)**. Among different voltammetry methods, SWV and DPV are often used due to their high sensitivity.[32] They have also been widely utilized for cancer biomarkers detection in numerous matrices. The voltammetric techniques have been used to detect numerous cancer biomarkers including, IL-10, HER2, Osteopontin (OPN), HEp-2, HT29, PSA, CA153, HCT, etc.[33,34]

Impedimetric techniques have also been proved to be a promising method for cancer biomarker detection due to their low excitation voltage, fast speed, and high sensitivity.[35,36] More importantly, they can be used for long-time, real time, and on-site detection.[37,38] Electrochemical impedance spectroscopy (EIS) is the most often used impedance method. Compared with voltammteric methods which need apply about −200 mV ∼ 600 mV excitation voltage, the EIS just need 5 mV or 10 mV excitation voltage.[39] The low excitation voltage makes it become a more undamaged detection technology. For bioelectrochemical analysis systems that require long-term monitoring, electrode heating is a problem that can cause changes in the biological microenvironment and damage to the electrodes. Low excitation voltage does not generate a lot of heat, hence EIS is more suitable for long-term and real-time detection. Furthermore, the EIS provides multiple parameters of the biosensing surface. Using a redox couple, typically a mixture of ferricyanide and ferrocyanide, the change in the charge transfer resistance (R_{et}) is obtained. Usually, the R_{et} is inversely proportional to the rate of electron transfer. The double layer capacitance (*CPE*) and the R_{et} describe dielectric and isolation features of electrode-electrolyte interface. The electrolyte resistance (R_s) and the Warburg impedance (Z_w) characterize the properties of an electrolytic solution and diffusion limitation for redox probe to reach the electrode surface and do not affect electron transfer at the electrode surface. The detection in the broad frequencies range (10^{-4}–10^6 Hz) makes the EIS strategy useful for diffusion analysis and for providing kinetics characteristics. Generally, at low frequencies (f < 1 mHz) the impedance is determined by the DC-conductivity of the electrolyte solution and at higher frequencies (f > 100 kHz), inductance of the electrochemical cell and connecting wires dominate the system.[40–42] For most cancer biomarkers, like cancer cells, protein biomarkers and nucleic acid biomarkers, can be detected by EIS. However, for some special biomarkers, like VOCs, are not suitable by choosing EIS as an electrochemical detection technology. Tang et al.[43] reported an EIS biosensor to detect MDA-MB-231 breast cancer with LOD of 10 cells mL^{-1}. Parekh and coworkers[38] developed an impedance-based non-invasive method for real-time analysis of MCF-7 (less aggressive) and MDA-MB-231 cells (more aggressive). Detailed slope-analysis of impedance curves at different growth phases showed that MDA-MB-231 had higher proliferation rate and intrinsic resistance to cell death which allowed to grow in nutrient and space limiting conditions. The analytical performance of other EIS-based biosensors for both molecule and cell biomarkers detection were listed in Table I.

Capacitive sensing is a type of non-faradaic impedance sensing. Arya et al.[44] developed a capacitive aptasensor to detect and estimate human epidermal growth factor receptor 2 (HER2), a biomarker for breast cancer, in undiluted serum. The aptasensor exhibited logarithmically detection of HER2 from 1 pM to 100 nM in both buffer and undiluted serum with limits of detection lower than 1 pM.

Conductometric sensing strategies provide exciting possibilities for the advanced development of new detection technique for bioanalytical applications, thanks to their unique advantages, e.g., being suitable for miniaturization and large-scale production, not requiring a reference electrode and only requiring a low driving voltage. Liang et al.[45] used conductometric immunoassay to detect alpha-fetoprotein (AFP) in sera of liver cancer patients. Under optimal conditions, the developed immunosensing system exhibited good conductometric responses toward target AFP within a dynamic linear range of 0.01–100 ng mL^{-1} at a relatively low detection limit of 4.8 pg mL^{-1}. Bhardwaj et al.[46] developed tetracyanoquinodimethane (TCNQ)-doped thin films of copper-MOF, $Cu_3(BTC)_2$ based conductometric immunosensing platform for the quantification of prostate cancer antigen, it provided PSA detection in a dynamic linear range of 0.1–100 ng mL^{-1} to attain a limit of detection at 0.06 ng mL^{-1}. Lin et al.[47] constructed silicon nanowire conductometric sensors for the detection of apolipoprotein A1, a biomarker for bladder cancer. The sensor has a dynamic range that covers the 0.2 ng mL^{-1} to 10 μg mL^{-1} concentration range and a detection limit of approximately 1 ng mL^{-1}.

Field-Effect Transistor (FET) The FET is a type of transistor that uses an electric field to control the conductivity of a channel (i.e. a region depleted of charge carriers) between two electrodes (i.e. the source and drain) in a semiconducting material. Control of the conductivity is achieved by varying the electric field potential, relative to the source and drain electrode, at a third electrode, known as the gate. Although there are many different types of FET devices, the current focus in biosensing applications is on ISFET (ion-selective field-effect transistor) and EnFET (enzyme field-effect transistor) devices. Modern complementary metal-oxide semiconductor (CMOS) manufacturing techniques provide advantages of miniaturization, parallel sensing (e.g. sensing arrays), and capabilities to be integrated with electronic circuits and systems. This would be a major advantage for solid-state based biosensors to compete with other bio-sensing mechanism in the future. Syu et al. reviewed the application of FET based biosensor in cancer detection.[48]

Bio-surface and receptor.—In general, biosensors are comprised of: (A) a bioreceptor (biosensing element), to which the analyte has a highly specific binding affinity; (B) a biosurface/biointerface architecture, which provides an environment for the proper functioning of the bioreceptor; (C) a transducer converting the physical phenomenon or chemical response resulting from the analyte's interaction with the biological element into readable signals (e.g., physicochemical, optical, piezoelectric, electrochemical, etc.). The latter can be reproducibly measured, quantified and processed; and, (D) an associated electronics comprising of signal amplifier, signal processor and an interface, like a display, which finally allows a user-friendly visualization and evaluation of the data.[49] For electrochemical biosensor, electrode is

Table I. Analytical performance of selected ultrasensitive EIS-based biosensors for both molecule and cell detection.

Cancer type	Receptor/ Biomarker	Sample	Device/indicator	Label/Label free	LOD	LR	Ref.
Breast	Antibody/UBE2C	MCF-7 cell extract	GCE-PANI/ $[Fe(CN)_6]^{3-/4-}$	Label free	7.907 pg mL^{-1}	500 pg mL^{-1} – 5 mg mL^{-1}	197
	Antibody/HER3	serum	CNT-SPCE/	Label free	2 fg mL^{-1}	2 – 14 fg mL^{-1}	198
	DNA/BRCA1	serum	GE/$[Fe(CN)_6]^{3-/4-}$	Label free	0.3 fM	1.0 fM – 10.0 pM	64
NR	DNA/miRNA-21	Serum	GCE-AuNPs@MoS_2/ $[Fe(CN)_6]^{3-/4-}$	Label	0.45 fM	10 fM – 1 nM	199
Prostate	Aptamer/PSA	Serum	IDE/$[Fe(CN)_6]^{3-/4-}$	Label free	0.51 ng mL^{-1}	0.5 ng mL^{-1} – 5000 ng mL^{-1}	200
Breast, non-small-cell lung carcinoma	Antibody/HER-3	Artificial serum	GE/$[Fe(CN)_6]^{3-/4-}$	Label free	NR	0.2 – 1.4 pg mL^{-1}	201
Colorectal	aptamer/CT26 cells	NR	SPE/$[Fe(CN)_6]^{3-/4-}$	Label to form sandwich	2 cells mL^{-1}	10 – 1.0 × 10^5 cells mL^{-1}	202
Gastric adenocarcinoma	boronic acid/AGS cells	NR	PGE/$[Fe(CN)_6]^{3-/4-}$	Label free	10 cells mL^{-1}	10 –10^6 cells mL^{-1}	203
Brain	Antibody/MDM2	mouse brain tissue homogenate	GE/$[Fe(CN)_6]^{3-/4-}$	Label free	1.3 pg mL^{-1}	1 pg mL^{-1} –10 ng mL^{-1}	181

CNT: carbon nanotube, GE: gold electrode, MDM2: Murine double minute 2, NR: not reported, PGE: pencil graphite electrode, PSA: prostate-specifc antigen, UBE2C: ubiquitin-conjugating enzymes 2C.

used as transducer (Fig. 1A). The receptor is the most important component for the design of an electrochemical biosensor, which involves antibodies, lectins, peptides, deoxyribonucleic acid (DNA), peptide nucleic acids (PNAs), aptamers, and molecularly imprinted polymers (MIPs).[50–52]

Antibodies, also known as immunoglobulins, are the immune system related proteins, which can selectively bind to antigens with a high binding constant in excess of 10^8 L mol^{-1}. The most important advantage of antibodies is the specificity and affinity of these probes to target analytes.[50] Antibody-based electrochemical biosensors, also called electrochemical immunosensors, is a type of most common biosensor for cancer protein biomarkers detection.[53–56]

Lectins are natural proteins of non-immune origin with specific binding affinity for the glycan moiety of glycolipids and glycoproteins. However, lectins exhibit low affinity (dissociation constant $K_d = 10^{-3}$–10^{-4} mol L^{-1}) while binding with carbohydrates compared to antigen-antibody interactions with K_d in the subnanomolar range.[57] The lectins render them as valuable recognition elements for biosensing of glycoprotein tumor markers.[58]

Peptides are short chains of amino acid monomers linked by amide bonds. Generally, peptides contain fewer than 20–30 amino acid residues, whereas proteins contain as many as 4000 residues. Compared to antibody, the peptides are more stable, easier to overcome harsh environments and more amenable to synthesize at the molecular level.[59] A specific peptide for sensitive analysis of PSA was reported by He and coworkers,[60] it showed a good selectivity to PSA when bovine serum albumin (BSA), hemoglobin (HB), CEA and human alphafetoprotein (AFP) were in the analyte solution.

DNA is a stable, low-cost and easily adaptable molecule, which is an excellent building block for the construction of electrochemical biosensor.[61,62] DNA based electrochemical biosensors were used to detect the cancer related gene mutations.[63] Cui and coworkers[64] developed an oligonucleotides based EIS DNA biosensor for breast cancer gene marker BRCA1 detection. The LOD was 0.3 fM, which was efficient enough to detect DNA mismatches.

PNAs is a type of non-natural nucleic acid analogue whose backbone is composed by N-(2-aminoethyl) glycine motifs linked via peptide bonds. PNAs exhibits chemical and thermal stability in conditions where DNA/RNA would undergo degradation.[65,66] The biophysical properties of PNAs make it an excellent candidate for use in biosensing applications, particularly when used as the bioreceptor.[67] Micro RNAs[68] and ctDNA[69] were detected by electrochemical PNA biosensor.

Aptamers are single-stranded DNA or RNA sequences that can specifically bind to analytes by folding into well-defined three-dimensional structures.[70] Compared to antibodies, they have advantages of rapid in vitro selection, cell-free chemical synthesis, low immunogenicity and superior tissue penetration.[71] Wu et al. reviewed aptamer as a type of bioreceptor for cancer diagnosis and therapy.[72]

MIPs also have been used as receptor in electrochemical biosensors. Compared with other bioreceptors, MIPs posess better chemical stability. MIP receptor films are usually prepared by electrochemical polymerization, since they have the advantages of (1) simple and fast preparation, (2) high adherence to the electrode surface, (3) easy control of the film thickness and morphology and (4) high reproducibility in different conductive materials.[73,74] Electropolymerization of 2-aminophenol based MIP has been used in a DPV biosensor for cancer antigen 15-3 (CA 15-3) detection.[75] The same research group also reported electrochemical MIP sensor for the detection of human epidermal growth factor receptor 2 (HER2).[76] However, their results showed that the two analytes interfere with each other. The specificity of MIPs needs to be improved.

Label, label-free and redox species.—There are many methods to generate electrochemical signal. It could be label or label free (Fig. 2A) by adding an electro-active indicator or no indicator. Besides non-faradaic EIS, most of electrochemical measurement methods require the use of electro-active indicator.[77] Of course, for an analyte that is electrochemically active by itself, it is not necessary to add an electro-active indicator. Redox ions such as ferricyanide/ferrocyanide ions ($Fe(CN)_6^{3-/4-}$) are usually dissolved in the sample solution as electro-active indicator to record the output signals of electrochemical biosensors. Other commonly used redox species are electroactive organic compounds such as quinones/hydroquinones (e.g., anthraquinone), anthracyclines (daunomycin, doxorubicin), viologens, phenothiazines (thionine, methylene blue, toluidine blue) and quinoxaline derivatives (echinomycin). Metal-based redox species include simple metal complexes such as $[Ru(NH_3)_6]^{3+/2+}$ [78], organic metal chelates $[M(L)_3]^{3+/2+}$ where M stands for Fe, Co, Os or Ru, and L for heterocyclic nitrogenous bidentate ligands such as 2,2'- bipyri-

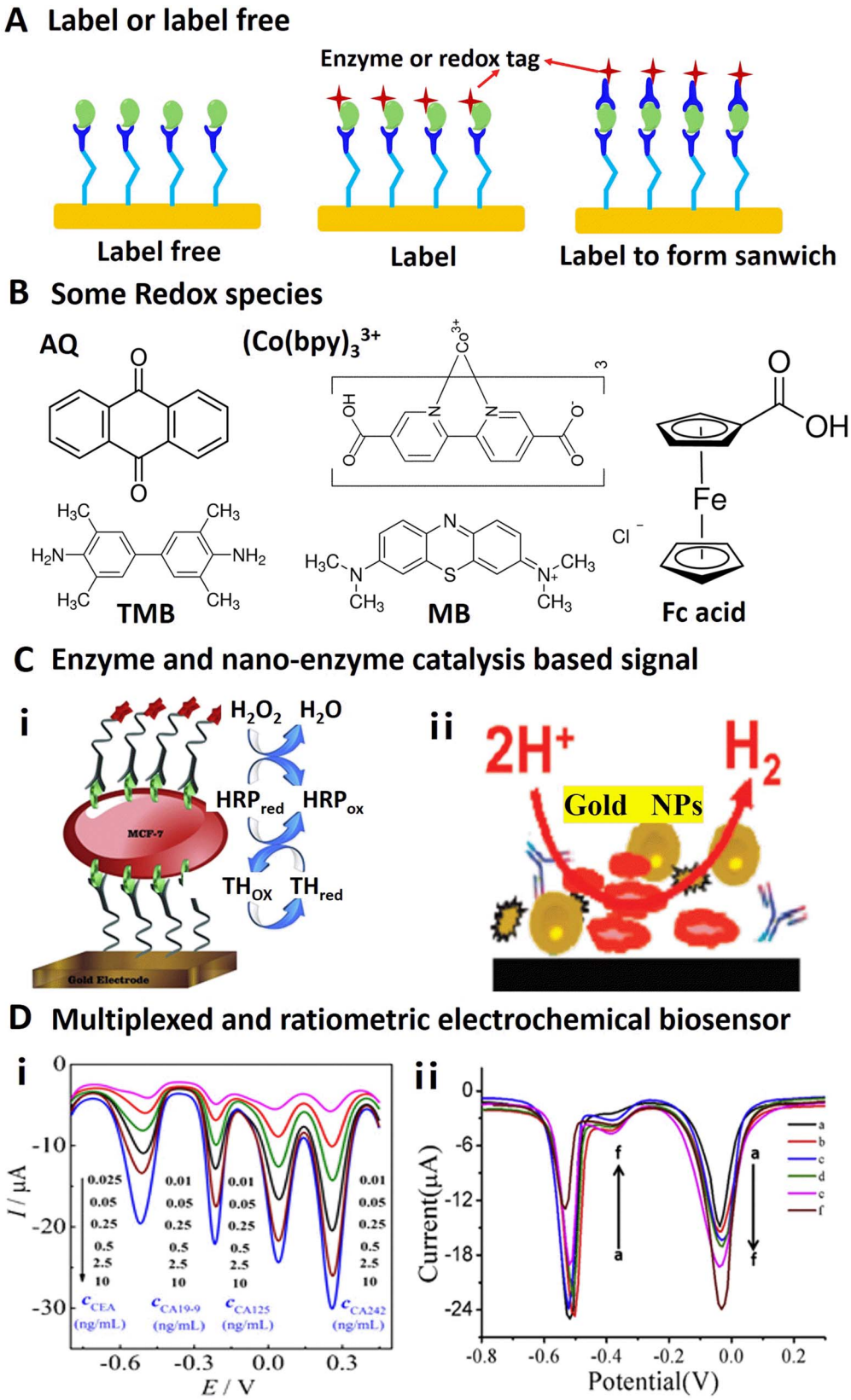

Figure 2. (A) Schematic diagram of label and label free style of electrochemical biosensors. (B) Some redox species. (C) Enzyme/nano-enzyme catalysis based electrochemical signal generator (reprint from Ref. 82, 84), (D) Signals of multiplex electrochemical assay (reprint from Ref. 86) and a ratiometric electrochemical biosensor (reprint from Ref. 87).

dine (bipy) or 1,10-phenantroline (phen), metalloporphyrins, oxoosmium(VI) complexes and metalloorganics such as ferrocene and its derivatives[79] (Fig. 2B). Heavy metal ions such as Cu^{2+}, Cd^{2+}, Pb^{2+} also used for electro-active indicators.[76]

Anthraquinone (AQ, potential at −0.37 V vs. Ag/AgCl) and methylene blue (MB, potential at −0.15 V vs. Ag/AgCl) were used by Liu et al.[80] to label aptamers, then the complexes were immobilizated on one electrode for simultaneous detection of interferon gamma (IFN-γ) and tumor necrosis factor alpha (TNF-α). Zhu et al.[81] applied AQ, thionine (TH), tris(2,2(-bipyridine-4,4(-dicarboxylic acid) cobalt(III) ($Co(bpy)_3^{3+}$) and ferrocenecarboxylic acid (Fc) as redox probes to simultaneously detect four biomarkers of colorectal carcinoma.[80] Among the known redox labels compatible with the DNA sensing platform, MB has emerged as the most versatile one, owing to its stability under a wide range of experimental conditions.

Catalytic reactions can be used as electro-active indicator. Both enzymes and nanoparticles with catalytic activity can produce electrochemical signals by catalyzing the reduction of hydrogen peroxide. The most commonly used enzymes in biosensing are glucose oxidase (GOx) and horseradish peroxidase (HRP), less commonly used enzymes comprise beta-lactamase, urea and urease.[24] Zhu et al.[82] used HRP labeled mucin 1 (MUC1)-binding aptamer to detection Michigan cancer foundation-7 (MCF-7) human breast cancer cells. Both hydrogen peroxide (H_2O_2) and thionine were used to the test solution, the occurrence of the HRP-catalyzed and thionine-mediated electrochemical reduction of H_2O_2 generated an apparent reduction (Fig. 2C-i). Saucedo et al.[83] used acid metabolites produced upon glucose induction by viable cells as redox probe. Escosura-Muniz et al.[84] applied catalytic effect of gold nanoparticles on hydrogen ion reduction as electrochemical signal output for biosensor (Fig. 2C-ii).

Conducting polymers, in particular, have been widely used in bioanalytical applications due to their inherent charge transport properties and biocompatibility and in biosensor applications owing to their specific sensitivity to very minor perturbations. Particularly the conducting polymer/CNT composites have been very much used in biosensors construction in recent years. Barsan et al. have published a related review paper.[85] Conducting polymer/noble metal composites are also used as electrochemical redox species which have great potential applications for multiplexed electrochemical biosensors. Wang et al.[86] synthesized four kinds of polyaniline derivative-Au/Pd composites as redox-active species to simultaneously detect four tumor biomarkers (CEA, carbohydrate antigen 19-9 (CA199), carbohydrate antigen 72–4 (CA72-4), and alpha fetoprotein (AFP). It is a good way to achieve electrochemically multiplexed bioassays (Fig. 2D-i).

Ratiometric signal transducing strategies can improve the precision of biosensor, which possess the unique merits of spatial-resolved signal readout and self-correcting toward possible false positive results (Fig. 2D-ii). Li et al. designed a ratiometric electrochemical immunosensor for nuclear matrix protein 22 (NMP22) based on bioinspired synthetic melanin nanospheres (SMNPs). SMNPs chelated with Pb^{2+} were used for the immobilization of captured primary anti-NMP22. And SMNPs chelated Cu^{2+} were employed to prepare signal labels after anchored with anti-NMP22 antibody. After sandwich-type immunoreaction, with the increasing concentration of NMP22, the stripping peak current of Pb^{2+} decreases while the stripping peak current of Cu^{2+} increases.[87]

Signal amplification.—Since the concentration of cancer biomarkers in body fluids are too low in early cancer stage, electrochemical signal amplification is needed. **Electrochemical ELISA** (enzyme-linked immunosorbent assay) is one of most reported methods for cancer protein biomarker detection. Conventional ELISA can barely reach less than nanomolar concentration level, which is inadequate to reach the clinical threshold of many protein biomarkers, so two main signal amplification strategies are used. One is modifying electrode with nanomaterials, the other is design second antibody-redox-enzyme complexes with nanomaterials.[6] Wang and Anzai[88] reviewed nanomaterials, including metal nanoparticles, carbon nanotubes, and graphene that have been used for fabricating electrochemical biosensors. Nanomaterials used to modify electrodes and used as signaling labels were both discussed. The electrodes modified with nanomaterials can increase the effective surface area of the electrodes and immobilize a large number of bioreceptors such as antibodies and nucleic acids. The nanomaterials used as signaling labels can also form sandwich structure and increase the output signals.[88,89]

2D nanomaterials and metal nanomaterials are most popular for electrode modification. Wang et al.[90] reviewed 2D nanomaterials based electrochemical biosensors for cancer diagnosis. Graphene and graphene oxide are the most popular 2D nanomaterial due to their excellent electrical performance. These two materials based electrochemical biosensor for early–stage cancer diagnosis has been reviewed by Balaji and Zhang.[91] Several gold nanomaterials based signaling labels are reported. Three main signal amplification mechanisms are used: electrostatic repulsion,[92] sterical hindrance,[93] and precipitation generation.[94] Zhao et al.[95] designed a cascaded signal amplification approach by using gold nanoparticle-$CaCO_3$ microspheres (AuNP-$CaCO_3$) to trigger pH responsive alginate hydrogel precipitation for sandwich-type impedimetric immunosensor. The hindrance effect of AuNP-$CaCO_3$ can significantly enhance the impedance response as the initial signal amplification. Then, part of $CaCO_3$ dissolves under weak acid conditions and releases Ca^{2+}, which can cross-link with alginate to generate an insoluble alginate hydrogel precipitate on the sensing interface, significantly increasing the impedance signal. They achieved a LOD of 0.09 fg mL^{-1} for PSA. The same group reported another signal amplification strategy by coupling cascade catalysis-initiated radical polymerization.[96] Copper-based metal-organic framework nanoparticles (Cu-MOF), as peroxidase mimics, combined with CA15-3 antibody (Ab_2) and glucose oxidase (GOx) were employed as immunoprobes to initiate radical polymerization by cascade catalysis. The method with an ultralow LOD of 5.06 μU/mL for CA15-3. Besides nanomaterials based signal amplification, electrochemical redox cycling amplification technology is also used for cancer biomarkers detection.[97] The schematic diagram of signal amplification mainly for electrochemical ELISA is showed in Fig 3A.

Rolling circle amplification (RCA) is a commonly used research tool for sensitive detection of DNA, RNA, DNA methylation, single nucleotide polymorphisms (SNP), small molecules, target proteins, and cancer cells.[98] The circular templates designed in RCA can make a single binding event amplified over a thousand-fold. The signal from a single binding event can be amplified in an exponential manner, so RCA is ideal for the required ultrasensitive detection. Liu and Luo et al.[99] reported padlock probe primer generating RCA for the detection of DNA methylation.

A DNA tetrahedron is utilized as a nanocarrier that can be immobilized on a gold electrode to generate RCA product to load hemin, an iron-containing porphyrin with chlorine, forming the G-quadruplex as a horseradish peroxidase like DNAzyme, which reduces methylene blue (MB) in the presence of H_2O_2 to yield a distinct current signal. The RCA based electrochemical biosensor can achieve a LOD as low as 0.1 fM. A representative schematic diagram is showed in Fig. 3B.[100] Electrochemical biosensor based on hybrid chain reaction (HCR),[101,102] nicking enzyme signaling amplification (NESA)[103] and enzyme-based exponential amplification[104] for cancer gene detection also have been reported.

Integration of electrochemical biosensor with microfluidic chips.—Microfluidic chips can grant electrochemical biosensor more powerful performance since it can integrate sample pretreatment process, conduct high-throughput multiplexed detection assay, and need little reagent.[105] Cancer molecular biomarkers and CTCs have been widely detected and analysized by electrochemical microfluidic devices.[106–109] Uliana and coworkers[110] developed a fully disposable microfluidic electrochemical array device (μFED) to detect the breast cancer biomarker estrogen receptor alpha (ERα). Array of 8 carbon-based working electrodes (8-WE), the pseudo-reference electrode (RE), and the counter electrode (CE) were constructed using a simple procedure based on the use of a cutter printer for rapid prototyping. The construction process of the microfluidic device was shown

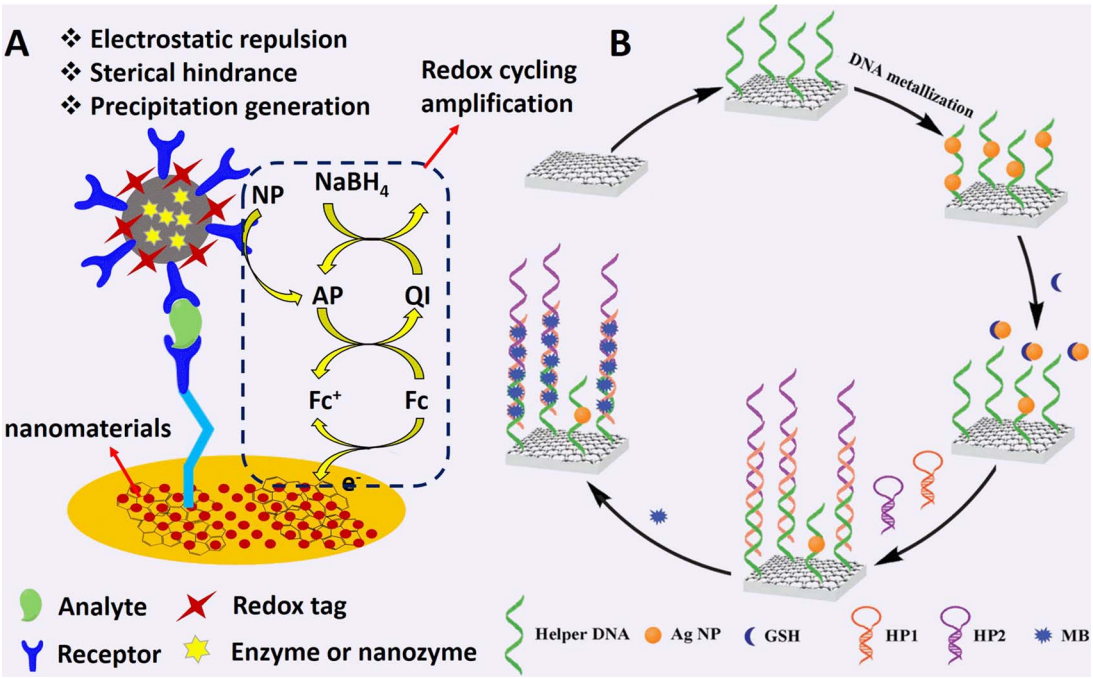

Figure 3. Schematic diagram of electrochemical signal amplification strategies. (A) is for electrochemical ELISA. (B) is for RCA based signal amplification (reprint from Ref. 100).

in Fig. 4A. The extremely simple and cost-effective μFED manufacturing process would enable the technique to be used in routine public health testing, making an important contribution to cancer diagnosis and personalized treatment of cancer. An inexpensive high-throughput electrochemical array featuring 32 individually addressable microelectrodes was reported Tang et al.[111] Highthroughput analyses were realized using eight 32-sensor immunoarrays connected to the miniaturized 8-port manifold, allowing 256 measurements in <1 h. This system was used to determine prostate cancer biomarker proteins PSA, prostate specific membrane antigen (PSMA), interleukin-6 (IL-6), and platelet factor-4 (PF-4) in serum. Wang and coworkers[112] designed a paper-based electrochemical aptasensor for simultaneous detection of CEA and NSE in clinical serum samples (Fig. 4B). The device was fabricated on four layers of cropped cellulose filter papers, and thus had the filtering function. For the operation of the device, a sample was introduced from the sample inlet located on the side, flowed through the microchannel, and reached the location where the three-electrode system was screen printed. Multiplexing analysis of the device was realized since the capillary action guided the sample to the two independent working electrodes on the paper within hot wax printed channels, and then recognized by the corresponding DNA aptamers. Hong et al.[107] fabricated an integrated microfluidic chip which can preconcentrate the methylated DNAs using ion concentration polarization (ICP) and electrochemically detect the pre-concentrated DNAs. The chip is divided into a preconcentration channel and a buffer channel independently and connects them through a Nafion pattern. Four pneumatic valves were introduced to carry out the process of transferring the pre-concentrated methylated DNA to the sensing chamber and the incubation in the chip. The chip consists of three layers, including the gold electrode with a Nafion pattern on the bottom, the pre-concentration layer on the second layer, and the valve layer on the top layer (Fig. 4C).

Cancer Biomarkers and Electrochemical Biosensor-Based Liquid Biopsy

The United States National Cancer Institute (NCI) defines a biomarker as "a biological molecule found in blood, other body fluids, or tissues that is a sign of a normal or abnormal process or of a condition or disease. A biomarker may be used to see how well the body responds to a treatment for a disease or condition". Cancer biomarkers can be of various molecular origins, including DNA (ie, specific mutation, translocation, amplification, and loss of heterozygosity), RNA, or protein/glycoproteins (ie, antigens, MMPs, cytokines). They can be found in body fluids such as blood, serum, urine, saliva, tears or cerebral spinal fluid[5].[113] Detection of cancer biomarkers can be clinically applied for early cancer diagnosis, accurate pretreatment staging, determination of response to chemotherapy treatment, and monitoring disease progression.[114] Lots of efforts have been made to identify biomarkers for specific cancers and use them to predict the risk. Protein and nucleic acid biomarkers associated with different cancers were list in Table II.

Protein biomarkers.—Extensive efforts have been devoted to developing ultrasensitive electrochemical biosensors for the detection of cancer protein biomarkers with high selectivity.[115,116] Cancer protein biomarkers include antigens, MMPs, and cytokines. The antigens include CEA, cancer antigen 15-3, 19-9, 125, PSA etc.[50,117] CEA is the most extensively used biomarkers for diagnosis and management of colorectal cancer, ovarian carcinoma and breast cancer. Many Electrochemical immunosensors based on monoclonal anti-CEA or polyclonal anti-CEA have been proposed for CEA detection.[118] A heterogeneous immunosensor was reported by Gu et al.[119] by immobilizing polyclonal anti-CEA onto Fc-labeled AuNPs on the gold electrode. By using SWV detection method, LOD was calculated to be 0.01 ng mL^{-1}. Tang et al. developed an electrochemical immunosensor by attaching CEA antibody (CEAAbs) onto glutathione (GSH) monolayer-modified AuNPs. Then these bioconjugates were fixed on a gold electrode by electro-copolymerisation with o-aminophenol (OAP). When CEA are bound with the electrode, the formed complexes increase the electron transfer resistance of the redox pair, resulting in signal change monitored by CV and EIS.[120] Similarly, Rusling et al. used GSH conjugated AuNPs and magnetic nanoparticles (MNPs) for the detection of PSA. PSA was bound by anti-PSA Abs conjugated to AuNPs of the electrode surface. Then PSA was sandwiched by multi-labelled MNPs bearing secondary PSA Abs. A microelectronic electrode gave a signal due to the reduction of H_2O_2 and allowed the detection of PSA.[121]

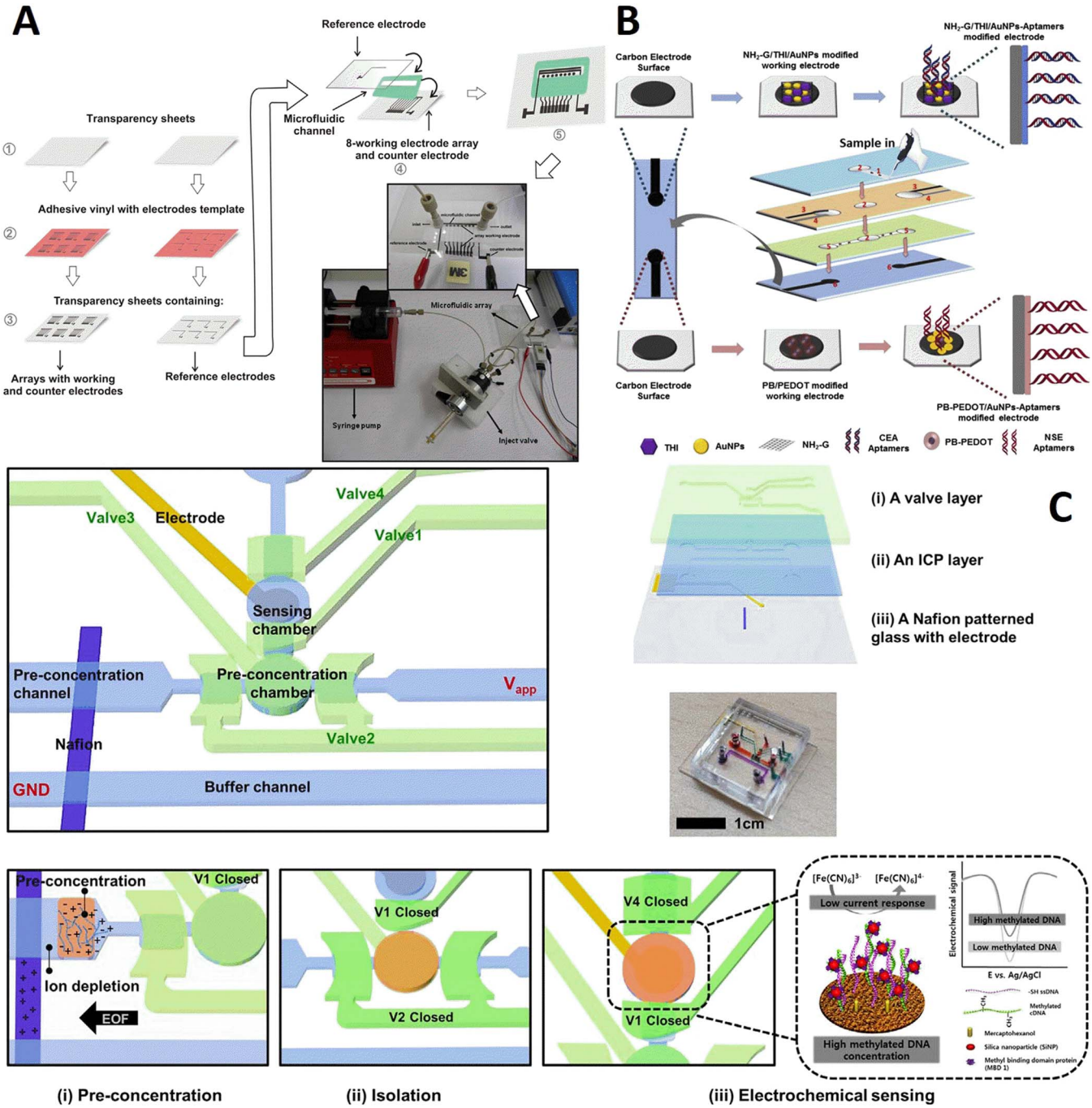

Figure 4. (A) Construction of the μFED and image of the microfluidic setup (reprint from Ref. 110). (B) Fabrication and modification process of the multi-parameter electrochemical paper-based aptasensor (reprint from Ref. 112). (C) Schematic diagram of the pre-concentration and electrochemical detection of the methylated DNA (reprint from Ref. 107).

Matrix metalloproteinases (MMPs) are considered to be related to the occurrence, development, invasion, and metastasis of cancers.[122] MMPs are capable of degrading a variety of component proteins in extracellular matrix (ECM),[123] such as collagen, elastin, gelatins, and casein.[124] The content of MMP-7 in serum samples is associated with lymph node metastasis in patients with some cancers, such as salivary gland cancer,[125] colon adenocarcinoma,[126] and high-grade renal cell carcinoma.[127] Cleavage strategy based electrochemical biosensors used for MMPs detection. Wei et[128] developed a peptide cleavage biosensor with LOD of 17.38 fg mL^{-1} by using MMP-7 as cleavage enzyme between leucine and alanine in peptides, causing the current signal change measured by SWV. Similarly, another study designed a dual-reaction triggered sensitivity amplified cleavage-based biosensor for MMP-7.[129] The proposed MMP-7 biosensor combined dual catalytic reactions and thus showed ultralow LOD of 3.1 fg mL^{-1}.

Cytokines are signature biomarkers of immune response to cancer. They are widely used to track disease progression and monitor treatment outcomes.[130] Several cytokines are associated with the development of multiple myeloma, leukemia, and breast cancer cell.[13,14] Interleukins (ILs) are one type of cytokines that were first discovered to be secreted by white blood cells. They can stimulate cancer cell proliferation and accelerate tumor progression. Serum interleukin-6 (IL-6) is associated with neck squamous cell carcinoma, colorectal, gastrointestinal and prostate cancers. The mean value of IL-6 is

Table II. Biomarkers associated with different cancers.

Cancer type	Protein	DNA	microRNA	Ref.
Lung	ANXA2, CEA, Chromogranin A, CA 19-9, CYFRA 21-1 (CA-19 fragment), NSE, NY-ESO-1, SCC, SAA1, HER1	P53, P16, Ras genes, Telomere length and telomere-related genes, EGFR gene (c-ErbB-1 and c-ErbB-2)	miR-21 (in sputum)	3,204–207,208
Breast	BRCA1, BRCA2, CEA, CA 15-3, CA 27, CA29, EGFR, EpCAM, HER2, MUC1	BRCA1, BRCA2	miR-21, miR-373, miR-182, miR-1246 and miR-105	4,5,33,209
Ovarian	CA 125 (MUC-16), CEA, EGFR4, Claudin-4			50,210
Prostate	PAP, PSA, Sarcosine	TEMPRSS2	miR-21, miR-141, miR-375	211,212
Gastric/Stomach	CA72-4, CA19-9, CEA, IL-6	PVT1	miR-21, miR-331, miR-421	213
Liver	AFP, CEA, DCP, GP73, GPC 3		miR-21, miR-122, miR-16	214,215
Bladder	BAT, BLCA4, CYFRA 21-1, FDP, NMP22, HA-Hase,	TP53, BCL2, CDKN2A, NID2	miR-200a, miR-126, miR-125b	216
Colorectal	CA 19-9, CEA, CCSP-2	P53, KRAS	miR-92a, miR-135b	217–219
Pancreas	CA 19-9, PAM4,		miR-21, miR-155, miR-196	56,220
Brain	MDM2			181

AFP: α-1-Fetoprotein, ANXA2: Annexin A2, CA: Cancer Antigen, CRP: C-reactive protein, EGFR: epidermal growth factor receptors, IL: Interleukin, HER2: Human epidermal growth factor receptor 2, KRAS: V-Ki-ras2 Kirsten rat sarcoma viral oncogene homolog, MDM2: Murine double minute 2, NSE: Neuron-Specific Enolase, P53: protein supressor gene, PAP: Prostatic acidic phosphatase, PSA: Prostate-Specific Antigen, SAA1: Serum amyloid A1, SCC: Squamous Cell Carcinoma antigen.

$\sim$ 20 pg mL^{-1} in cancer patient compared to $\sim$ 6 pg mL^{-1} in healthy individuals.[131] Munge et al. used GSH modified AuNPs to detect IL-6 in serum. GSH-AuNPs were immobilized onto pyrolytic graphite and polydiallylammonium bromide. Bovine serum albumin (BSA) and IL-6 Abs were then attached to GSH-AuNPs. IL-6 bound Abs was then sandwiched with secondary Abs labelled by HRP, streptavidin and biotin. This complex enhanced the reduction of H_2O_2 and thus the emitted signal allowed the detection of IL-6.[132] Other cytokines (including growth factors), like IL-1β, IL8, tumor necrosis factor-alpha (TNF-α), human epidermal growth factor receptor 2 (HER2), platelet-derived growth factor (PDGF), vascular endothelial growth factor (VEGF) were also reported as cancer biomarkers.[33,50,133]

Nucleic acid.—Cell-free nucleic acids (cfNAs), such as cfDNA and cfRNA, are present at significant levels in the blood of cancer patients, are highly desirable targets for electrochemical sensor. A fraction of hematogenous cfDNA originates from tumors are termed as **circulating tumor DNA (ctDNA)**. Traditionally, cancer related ctDNA have been detected using a variety of untargeted methods such as digital karyotyping, personalized analysis of rearranged ends (PARE), whole-genome sequencing of ctDNA, and targeted approaches such as conventional and digital PCR-based methods and deep sequencing-based technologies.[134,135] Recent advances of electrochemical biosensor based liquid biopsy technic will potentially lead to the development of diagnosis, prognosis, therapy response monitoring, and tracking the rise of new mutant sub-clones in cancer patients. Compared to traditional tumor tissue biopsies, electrochemical sensor can directly detect cancer-implicated mutant ctDNA without the need for invasive sampling of tissue. An electrochemical sensing platform was recently reported for cfDNA detection which based on nitrogen-doped multiple graphene aerogel/gold nanostar.[136] The DPV signal linearly increases with the increase of cfDNA concentration in the range from 1.0×10^{-21} g mL^{-1} to 1.0×10^{-16} g mL^{-1} with the detection limit of 3.9×10^{-22} g mL^{-1}. Chen et al.[137] reported electrochemical interrogation of circulating methylated DNA from plasma samples using a paired-end tagging. In the method, the first two DNA primers each labeled with digoxigenin (Dig) and biotin at the 5′ end were designed to discriminate methylated DNA from the unmethylated one and its subsequent amplification. The resulting paired-end tagged amplicons were incorporated on the electrode covered with anti-Dig. This modification on the electrode enhanced the efficiency of target capturing by avoiding biotin from the interface. Then, avidin-HRP molecules were captured by the biotinylated amplicons. The method was able to detect as little as 40 pg of methylated genomic DNA and as low as 1% methylation level. The clinical sensitivity was 91% (10/11 patient samples). In a similar way, this sequential discrimination–amplification strategy was used for circulating methylated DNA detection with single-copy sensitivity.[138] Firstly, a nanostructured self-assembled tetrahedral DNA was immobilized on the surface of a gold electrode to capture amplicons. The tetrahedral DNA probes notably increase target hybridization and reduce the non-specific adsorption of amplification by-products owing to the inflexible scaffold, ordered orientation, and well controlled spacing. The hybridization event resulted in a measurable electrochemical signal through an enzymatic amplification strategy. Using this sensitive detection technique, interrogation of as few as one methylated DNA molecule in the presence of a 1000-fold excess of unmethylated alleles was achieved. Apart from amplification-based strategies, several other novel approaches have been devised for direct and ultrasensitive detection of ctDNA. For example, Das et al.[139] designed an electrochemical clamp assay that can directly detect mutant circulating nucleic acids in patients' serum. By using an assortment of oligonucleotides, selective binding of mutated sequences to a chip-based electrochemical sensor is facilitated. It can detect DNA mutations within 15 minutes with high sensitivity and specificity.

MicroRNAs (miRNAs) are a class of small, non-coding endogenous RNAs with about 22 nucleotides long that can negatively control their target gene expression at post-transcription levels. A large group of miRNAs categorized as "circulating miRNA" are released into the blood and their expression level is specifically related with tumorigenesis, cancer development and metastasis.[140,141] There are numerous methods for miRNA detection and some of them including real-time qPCR, northern blotting, microarray, and deep sequencing of transcriptome (RNAseq).[140,142] The voltammetric methods are the most frequently used electrochemical techniques for qualitative and quantitative analyses of miRNA. In electrochemical miRNA biosensors, cyclic voltammetry (CV), differential pulse voltammetry (DPV), and square wave voltammetry (SWV) are the most attractive techniques due to higher sensitivity.[143–145] Kaplan et al.[146] developed biosensor based on electropolymerized polypyrrole (PPy) modified pencil graphite electrodes (PPy/PGE) to detect the miR-21 from breast cancer cell line MCF-7. EIS was used as the readout technics and they achieved a LOD of 0.17 nM.

Other molecular biomarkers.—Besides protein and nucleic acid biomarkers, other molecules can also be biomarkers for cancers. Volatile organic compounds (VOCs) that are emitted from cancer cell metabolism are considered to be important indicators for biochemical processes occurring in cancer cells. Analysis of VOCs may be capable of prognosticating and diagnosing early cancer.[147,148] Zhang et al.[149] developed an electrochemical biosensor with Au-Ag alloy composites-coated MWCNTs sensing interface for ultrasensitive detection of volatile biomarkers in MGC-803 gastric cancer cells. CV studies showed that the fabricated sensor could detect 3-octanone in the range of 0∼0.0025% (v/v) and a detection limitation of 0.3 ppb. The sensor could also detect butanone in the range of 0 ∼ 0.055% (v/v) with a detection limitation of 0.5 ppb. Good selectivity was exhibited by the sensor as well.

Exosomes.—Cancer-derived exosomes have drawn much attention for the early diagnosis and drug sensitivity analysis of cancer, which typically have sizes of 30–150 nm and densities of 1.13–1.19 g mL^{-1}. They play critical roles in tumorigenesis and progression, including immunosuppression, angiogenesis, cell migration, invasion, and more importantly, carring information of the tumor microenvironment. They can be obtained from body fluids (such as serum, plasma, and urine) with high abundance and stability.[150–153] Jeong and coworkers reported a portable magneto− electrochemical device and applied it to screen extracellular vesicles in plasma samples from ovarian cancer patients. Four different biomarkers (CD63, EpCAM, CD24, and CA125) along with their respective IgG controls can be simultaneously measured. By combining magnetic enrichment and enzymatic amplification, the device was able to detect exosomes at a sensitivity of <105 vesicles.[154] Li et al.[155] used EIS to quantify both external (tetraspanin) and internal (syntenin) exosome-specific markers with LOD of 1.9×10^5 particles mL^{-1}. Wang and colleagues[156] developed a DNA nanotetrahedron (NTH)-assisted aptasensor exhibiting a 100-fold increase in sensitivity compared with traditional aptamer-based biosensors. The principle of NTH enhancement allows for the maintenance of spatial orientation and reduces the hindrance effect for better biomolecular recognition, the LOD was calculated to be 2.09×10^4 exosomes mL^{-1}.

Cancer Cell Detection, Analysis and Monitoring

Cancer cell detection.—Sensitive detection of cancer cells can provide effective information of progression of cancer. Cancer cell of particular type can be specifically detected by electrochemical biosensors. Li et al.[157] proposed an efficient electrochemical method to detect and distinguish breast cancer cells via recognition of two tumor markers human mucin-1(MUC1) and CEA expressed on the surface of breast cancer cell MCF-7. The breast cancer cells MCF-7 are first recognized due to the specific interaction between MUC1 on the cell surface and its aptamer molecules, which are immobilized on a gold electrode surface. Subsequently, another tumor marker CEA on the cell surface is captured by CdS nanoparticles (CdS NPs) labeled anti-CEA. A label free aptasensor was reported by Wang et al.[158] for detecting cancer cell constructed through layer by layer assembly technique (LBL) with ferrocene-appended poly(allylamine hydrochloride) functionalized graphene (Fc-PAH-G), poly(sodium-p-styrenesulfonate) (PSS) and aptamer (AS1411). Differential pulse voltammetry (DPV) was performed to investigate the electrochemical detection of HeLa cells. Layer by layer construction of electrode helps improve the sensitivity and enhance amplified signal. A wide LR from 10 to 10^6 cells mL^{-1} with a detection limit as low as 10 cells mL^{-1} was achieved. Seenivasan et al.[159] used a dual-working electrode (WE) surface to improve the specificity of the detection system. One of the WE surface is functionalized with anti-Melanocortin 1 Receptor antibodies specific to melanoma cancer cells, while the other WE act as a control (i.e., without antibody), for detecting non-specific interactions between cells and the electrode. The method is described and shown to provide effective detection of melanoma cells at concentrations ranging between 25 and 300 cells per 20 mL after a 5 min incubation and 15 s of DPV measurements. The estimated limit of detection was ∼17 cells. In recent years, selectively capturing and isolating cancer cells have drawn more attention. Sun et al.[160] uses an electrochemical desorption method for the efficient release of HepG2 tumor cells by cleaving the Au-S bonds on the SPGE interfaces. The microscope image demonstrated that the released cells still maintain high viability for further study. Instead of using bioreceptor to recognize cell, some researchers use electric cell-substrate impedance sensing (ECIS) to achieve dynamic, real-time, non-invasive, and label-free monitoring of cell growth and viability. Pan et al.[161] reported a 3D electric cell/matrigel-substrate impedance sensing (3D ECMIS) platform for human hepatoma cells (HepG2). Based on this 3D ECMIS sensor, it is convenient to monitor the cell growth and viability and provide more dynamic data for drug screening.

Circulating tumor cells (CTCs) are cells released from the primary tumor into the bloodstream. CTCs as a prognosis factor and their use for early detection of metastasis events are recognised for several tumors such as breast, colorectal, lung and prostate cancers.[162] Several electrochemical biosensors have been developed for CTCs detection. DPV biosensor integrated in a microfabricated glass chip was employed by Moscovici and coworkers for rapidly detecting circulating prostate cancer cells. The binding of a prostate cancer cell onto the anti-EpCAM antibodies - modified gold surface alters the interfacial electron transfer of $[Fe(CN)_6]^{3-/4-}$. The biosensor allows high-efficiency readout of small cell populations within 15 min.[163] Xiang[164] synthesized a new aptamer-functionalized and gold nanoparticle (AuNP) array-decorated magnetic graphene nanosheet recognition probe to detect CTC. The incubation of the probes with the sample solutions containing the target CTCs can lead to efficient separation of the CTCs and result in the generation of two distinct voltammetric peaks by SWV. This work showed a detection limit as low as 4 and 3 cells mL^{-1}. After capturing and detecting, release of cancer cell from substrate without damage plays key role in subsequent study of cell analysis. Shen et al.[165] combined rolling circle amplification (RCA) with antiepithelial cell adhesion molecule (EpCAM) antibody-modified magnetic nanospheres for highly efficient capture and ultrasensitive detection of MCF-7 in peripheral blood. They achieved a LOD of 1 cells mL^{-1}.

Compared to cell population analysis, single-cell analysis suffers from some challenges, such as the small size of a single cell and the low expression levels of biomolecules at single-cell levels. Wang et al.[166] demonstrated the fabrication of 3D micro/nano structured graphene interface and their outstanding capability of impedance signal sensing in single cancer cell. Compared to classic 2D gold interface, the 3D graphene biointerface significantly improves the capture efficiency and the sensing sensitivity of single cell, demonstrating the increment in impedance signal of about 100% at the nodes of cell state change. Sanchez et al.[167] reported an electrochemical pulse amperometric detection based device for genetic profiling of single CTC. A panel of seven markers were analyzed owing to their high prognostic value for breast cancer tumors: CD24, CD44, CDH1, CDH2, ERBB2, HUWE1, and KRT19.

Phenotypic analysis.—Phenotypic profiling of cancer cells can reveal vital tumor biology information. The expression of CD44 is demonstrated to be related with the tumorigenic and metastatic potential of Breast Cancer Stem Cells (BCSCs), which is also a potential therapeutic target in treatment of breast cancer. Zhao et al.[168] used linear sweep voltammetry (LSV) to detect the CD44 on cancer cells with a limit of detection of 2.17 pg mL^{-1}. Ju's group[169] developed an electrochemical biosensor to detect glycan on the surface of K562 cells by using ferrocene–concanavalin A (Fc-ConA) as probe and bioreceptor. The results showed that the average number of mannose group on single living K562 cell was determined to be 3.0×10^{10}. Zhu's group[170] used a microfluidic chip that integrated electrochemical biosensors profiling the cell glycan expression in response to drugs. Both EIS and optical microscope technique were applied to detect the dynamic variation of glycan expression of K562 cell surface. Cluster of differentiation 147 (CD147), also known as extracellular matrix metalloproteinase inducer (EMMPRIN), plays an

essential role in tumor progression and metastasis, it can be found on the surfaces of 90% of micrometastatic tumor cells with high expression, indicating a poor prognosis for cancers. Zheng et al.[171] reported an assay to measure CD147/EMMPRIN expression on cancer cell surfaces by using electrochemical technique with a sandwich format. The CD147/EMMPRIN expressed on a single breast cancer cell can be calculated as 2.57×10^4 molecules $cell^{-1}$.

Cancer cell status and drug sensitivity monitoring.—Cancer cell analysis has been popular in recent decade, especially single cell analysis. Moreover, non-invasive methods to maintain the investigated cell for time dependent monitoring has been widely studied because of its importance in some crucial cases such as drug resistance in cancers.[172]

A bioelectronic sensor for monitoring the effect of anticancer drugs on single breast cancer cells was reported by Gharooni and Abdolahad.[173] Treating the cell by depolymerizing (MBZ) and Polymerizing (PTX) drugs induced different changes in impedance response of the sensor which exhibited a well correlation with biological effect of such drugs on the cells. Asphahani et al.[174] demonstrated the feasibility of a cell-based electrochemical biosensor that measures changes in cell impedance in response to analyte. These cell-based biosensors are commonly referred to as cytosensors and live cells are used as the biological sensing element in order to monitor changes induced by various stimuli. This type of sensor can be used to monitor the effects of anticancer agents on their target molecules.

As a primary modulator of cell microenvironmental reactions, the intracellular redox state plays significant roles in regulating cell behavior and function. The imbalance of intracellular redox homeostasis can affect DNA and RNA synthesis, signal transduction, enzyme activation and even cell proliferation, and may cause severe diseases, such as cancer and Parkinson's disease.[175,176] GSH, the most abundant thiol-containing tripeptide in biological systems, is regarded as the key molecule to maintain the intracellular redox homeostasis.[177] Hence, the measurement of GSH concentration has been used for evaluating the intracellular redox state. With the electro-active nature of GSH, its electrochemical detection via direct oxidation has been demonstrated in clinical application. Unfortunately, the direct oxidation of GSH using an electrochemical method is not a fully satisfactory strategy due to the poor voltammetric behavior of GSH.[178–180] Liu et al. developed a label-free ratiometric electrochemical biosensor to detect GSH in cancer cell lines (A549 and HeLa) and one normal cell line (NIH3T3). The result revealed that electrochemical signal was greatly increased upon mixing with the cancer cell (A549 and HeLa) extracts, while it exhibited a nearly negligible response after incubating with the normal cell (NIH3T3) extract. So the prepared biosensor could not only determine GSH in cell extracts but also discriminate between cancer and normal cells.[100]

Cancer Tissue/Cancer-On-A-Chip/Cancer Animal Model Detection and Analysis

Tissue/organ level and tumor microenvironment measurement.—Electrochemical biosensors have also been used for tumor tissue detection. Elshafey et al.[181] used EIS biosensor to detect MDM2 protein biomarker from mouse brain tissue homogenate. Torrente-Rodríguez and coworkers[182] developed amperometric biosensor to detect miRNAs in breast cancer tissue which were collected from breast cancer patients. Both of them are used to conduct procedures for cancer tissues digestion and analytes extraction. However, they are both in an invasive way.

In recent years, there has been a growing interest in the application of novel biosensors in tissue engineering or tissue and organ which are fabricated by 3D bioprinting in a non-invasive style. For example, on-site, real-time, continuously detection of small molecules such as glucose, lactose, and H_2O_2; as well as proteins of large molecular size, such as albumin and alpha-fetoprotein; and inflammatory cytokines, such as IFN-g and TNF-α.[183] Developing implantable, long-term stable, and biocompatible electrochemical biosensors to detect biomarkers or metabolites in tissue or organ is needed. It will greatly promote the establishment of in vitro models of cancer tissues or organs. Pettersen and coworkers[184–186] proposed the concept of the Sensing cell Culture Flask (SCCF) which provides a technological platform for cell culture monitoring without the need to deviate from tissue culture flasks (Fig. 5A). The SCCF platform allows the integration of ideally any electrochemical biosensor by simple adjustments during the post-processing steps of the chip fabrication. Weltin et al.[187] developed a multiparametric electrochemical biosensor based system for the dynamic online monitoring of human tumor microenvironment and cancer cell metabolism. The pH, oxygen, glucose, and lactate were measured with linear, long-term stable, selective and reversible behavior in the desired range (Fig. 5B).

Electrochemical biosensor integrated in organ-on-a-chip.—Despite the increasing number of organs-on-a-chip that have been developed in the past decade, limited efforts have been made to integrate a sensing system for in situ continual measurements of biomarkers from the three-dimensional (3D) tissues. Ortega et al.[188] present a custom-made integrated platform for muscle cell stimulation under fluidic conditions connected with a multiplexed high-sensitivity electrochemical sensing system for in situ monitoring. The detection sensitivity of the IL-6 and TNF-α are in the ng/mL (Fig. 5C). Khademhosseini research group designed microfluidic aptamer-based[189] and antibody-based[190] electrochemical biosensing platform for monitoring cardiac biomarker creatine kinase (Fig. 5D/E). They also reported an automated microfluidic platform of bead-based electrochemical immunosensor integrated with bioreactor for continual monitoring of Transferrin and albumin.[191] The capability of in situ continual monitoring of organ behaviors and their responses to pharmaceutical compounds over extended periods of time is critical in understanding the dynamics of drug effects and therefore accurate prediction of human organ reactions.

Animal level and in vivo detection.—In vivo electrochemistry has been a major tool to understand and reveal physiology and pathological processes from the molecular basis of living systems.[192,193] Griveau et al.[194] developed an electrochemical sensor that allows the in vivo detection of nitric oxide (NO) in tumor-bearing mice. It could be applied to the in vivo study of candidate anticancer drugs acting on the NO pathways. As a key marker for redox status of cells, GSH concentrations in cancerous cells are much higher compared to healthy cells.[195] Fierro et al.[196] used needle-shaped boron doped diamond (BBD) microelectrode as work electrode, silver wire as counter electrode and Ag/AgCl wire as reference electrode to detection the GSH concentration in subcutaneous xenograft tumors derived from human cancer cells in immunodeficient mice. Chronoamperometry measurement was mainly used to record the signal. The results showed that it is possible to measure in vivo the difference in concentration of GSH between cancerous and healthy tissues with a high reproducibility.

Conclusions and A Look to the Future

In this review, new advancements of electrochemical biosensor used in cancer liquid biopsy (include protein, nucleic acid, exosomes and cancer cell biomarkers detection), cancer drug sensitivity monitoring, cancer-on-a-chip and cancer in vivo detection were summarized and discussed. Though devices based on electrochemical biosensors rarely have been used in clinical patients, they have shown great potential and achieved great progress. Up to present, the LOD for molecular (protein, DNA miRNA,) biomarker and cancer cell detection based on electrochemical biosensor can be fg mL^{-1} and 1 cell mL^{-1} magnitude respectively (Table III). It is sensitive enough to detect these biomarkers in terms LOD (the normal level of some biomarkers was listed in Ref. 33,50). Among various electrochemical measurements, DPV and SWV provide a general and sensitive approach for direct or multistep measurement for biomarker detection. These methods also provide the possibility of simultaneous multi-analyte analysis in a short period of time. EIS provides the advantages of long-term, real-time and dynamic monitoring of cancer biomarkers and cells.

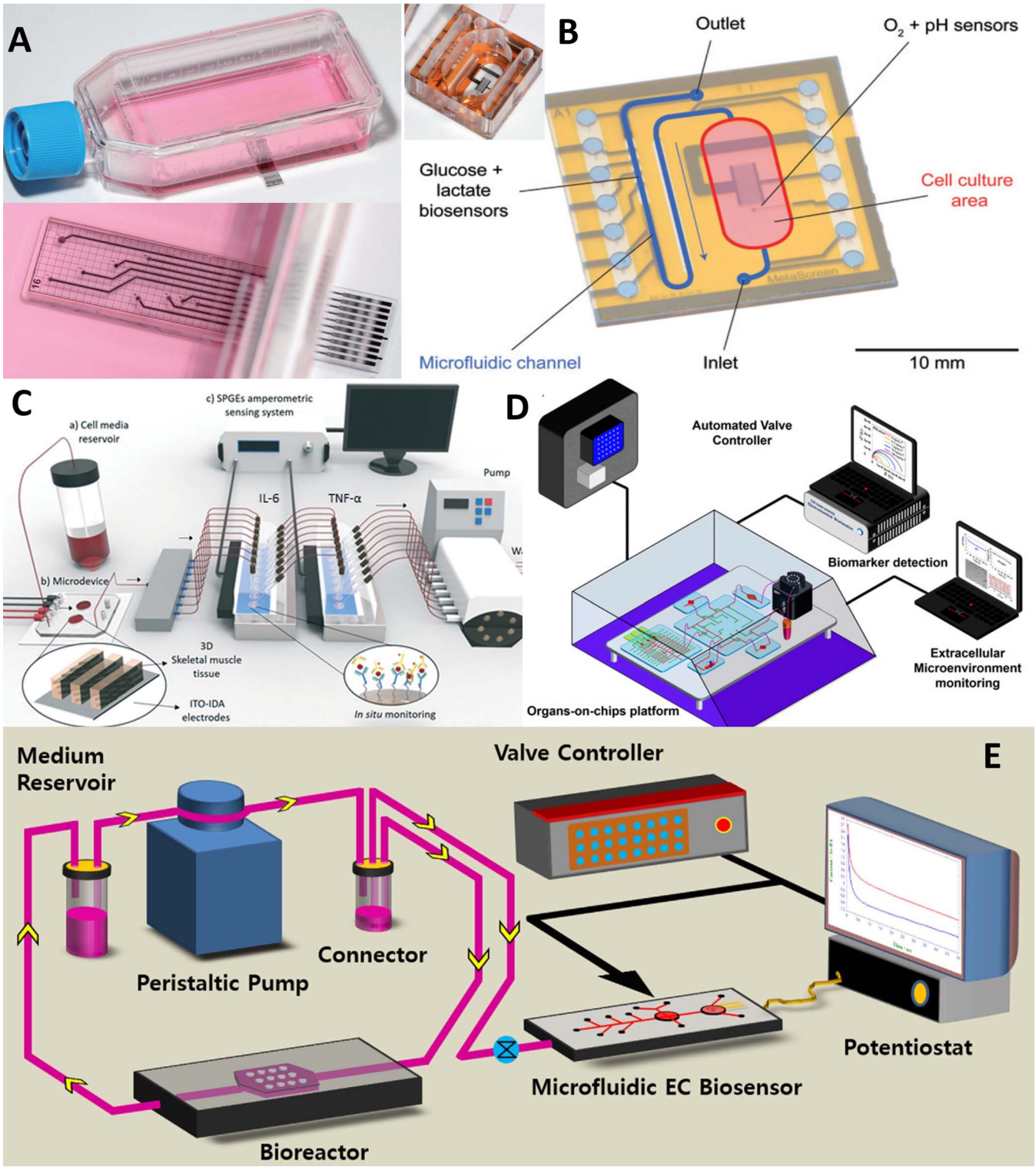

Figure 5. Electrochemical biosensor detection in a tissue level and integrated in Organ-on-a-chip. (A) Sensing Cell Culture Flask: a transparent sensor chip with different electrochemical sensors is embedded in the bottom of a tissue culture flask (reprint from Ref. 184–186). (B) Glassbased system for in situ amperometric oxygen sensors, potentiometric pH sensors and an interdigital electrode, as well as amperometric glucose and lactate biosensors integrated in the downstream microfluidic (reprint from Ref. 187). (C) Schematic overview of the configuration and function of the muscle-on-a-chip (reprint from Ref. 188). (D) Schematic of a full system where the multiorgan-on-a-chip platform was encased in an in-house designed benchtop incubator, and physical sensors, potentiostat for measuring electrochemical signals (reprint from Ref. 190). (E) Schematic of the microfluidic sensing system integrated with organs-on-chip for continual monitoring of a target biomarker (reprint from Ref. 189).

Table III. Analytical performance of selected ultrasensitive electrochemical sensors for both molecule and cell detection

Cancer type	Receptor/ Biomarker	Sample	Device/indicator	Label/Label free	Detection methods	LOD	LR	Ref.
Lung	Antibody/CEA	Serum	AgNPs/THI/ICP fibres GEC/thionine	Label free	DPV	0.5 fg mL^{-1}	50 fg mL^{-1} – 100 ng mL^{-1}	221
	Antibody/NSE	Serum	Conductive hydrogel -AuNPsGCE/ $[Fe(CN)_6]^{3-/4-}$	Label free	SWV	0.26 pg mL^{-1}	1 pg mL^{-1} – 200 ng mL^{-1}	222
Breast	DNA/ERα	MCF-7 cell lysate	μFED/HRP-H_2O_2-HQ	Label to form sandwich	Amperometry	10.0 fg mL^{-1}	16.6 to 513.3 fg mL^{-1}	110
	Aptamer/HER2	Serum	GCE/MB	Label free	DPV	0.21 ng ml^{-1}	0.5–2 ng ml^{-1} and 2–75 ng ml^{-1}	223
	Aptamer/MCF-7	Serum	GE/$[Fe(CN)_6]^{3-/4-}$	Label to form sandwich	DPV	8 cells mL^{-1}	10–10^5 cells mL^{-1}	224
	Anti-EpCAM-MNs/MCF-7	Peripheral blood	MGE/molybdate	Label to form sandwich	SWV	1 cell mL^{-1}	5–3 × 10^4 cells mL^{-1}	165
	Complementary RNA/miR-32 and miR-122	Serum	GNPS-SPEC/ $K_3[Fe(CN)_6]$ and $[Ru(NH_3)_6]Cl_3$	Label	SWV	5 aM	10 aM –1 μM	225
	Complementary oligonucleotide/ miRNA-155	Serum	Au-SPE/$[Fe(CN)_6]^{3-/4-}$	Label free	SWV	5.7 aM	10 aM – 1.0 nM	226
Colorectal stomach pancreas	Antibody/CA19-9	Serum	Au-PGO-GCE/ HRP-H_2O_2-Thi	Label to form sandwich	DPV	0.006 U mL^{-1}	0.015 – 150 U mL^{-1}	227
Stomach	Antibody/CA72-4	Serum	rGO-TEPA/GCE/ PtPd-Fe_3O_4- H_2O_2	Label to form sandwich	Amperometry	0.3 mU mL^{-1}	0.001–10 U mL^{-1}	228
Liver	DNA-aptamer/ HepG2 cells	Blood	SPGE/HRP-H_2O_2-HQ		DPV and EIS	5 cells mL^{-1}	10 – 1 × 10^6 cells mL^{-1}	160
	Antibody/AFP	Serum	Cu_2O@GO-GCE/TB	Label free	SWV	0.1 fg mL^{-1}	0.001 pg mL^{-1} – 100 ng mL^{-1}	229
	Antibody/AFP	Serum	Au-APTES-MCS/$[Fe(CN)_6]^{3-}$	Label to form sandwich	DPV	0.13 pg mL^{-1}	0.4 pg mL^{-1} – 100 ng mL^{-1}	230
Prostate	Antibody/PSA	Serum	GE/Cu^{2+}	Label to form sandwich	DPSV	0.020 ± 0.001 fg mL^{-1}	0.05 – 500 fg mL^{-1}	231
	Antibody/PSA	Serum	Nafion-rGO-CHO-MP-GE/$[Fe(CN)_6]^{3-/4-}$	Label free	DPV	1.6 pg mL^{-1}	5 pg mL^{-1} – 90 ng mL^{-1}	232
	Antibody/PSA	Serum	COOH-AgPtPd-NH_2-rGO/$[Fe(CN)_6]^{3-/4-}$	Label free	DPV	4 fg mL^{-1}	4 fg mL^{-1} – 300 ng mL^{-1}	233
	Anti-EpCAM-MNs/VCaP cells	Whole blood	VV chip/p-aminophenyl phosphate	Label to form sandwich	DPV	1 cell mL^{-1}		234

AFP: alpha fetoprotein, Anti-EpCAM: antiepithelial cell adhesion molecule, Au-APTES-MCS: Au NPs supported microporous carbon spheres functionalized by 3-aminopropyltriethoxysilane, Au-PGO: Au nanoparticles functionalized porous graphene, CA19-9: Carbohydrate antigen19-9, DPV: differential pulse voltammetry, DPSV: differential pulse stripping voltammetry, EIS: electrochemical impedance spectroscopy, ERα: estrogen receptor alpha, μFED: fully disposable microfluidic electrochemical array device, GCE: glassy carbon electrode, GE: gold electrode, GNPs-SPCE: gold nanoparticles-modified screen-printed carbon electrode, GO: graphene oxide, HER2: human epidermal growth factor receptor 2 protein, HQ: hydroquinone, SPGE: screen-printed gold electrode, hTERT: human telomerase reverse transcriptase, MB: methylene blue, MNs: magnetic nanospheres, Nafion-rGO-CHO-MP: Nafion/reduced graphene oxide/aldehyde methyl pyridine composite, NSE: neuron-specific enolase, SPCE: screen-printed carbon electrode, TB: toluidine blue.

Although electrochemical biosensors have achieved very low LOD, they usually relay on the multi-steps label strategies which make the experimental operation complicated. Developing ultrasensitive label-free electrochemical methods will be great potential in future works. Another challenge is that few portable electrochemical devices are in clinical usage owing to its limited accuracy and reliability. Hence, robust POC devices based on electrochemical biosensor are needed. Researchers should train the electrochemical biosensor with a large number of clinical samples to solve its reliability problems. For in vivo detection, developing wireless micro/nano electrochemical biosensor device will be ideal choice since it can work in a minimally invasive style.

Acknowledgments

This work was supported by National Science Foundation (CBET-1805514) to HSZ.

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